

The determinants of commercial banking profitability in low-, middle-, and high-income countries

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Highlights

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We analyze how bank-specific, macroeconomic and industry-specific factors affect the profitability of commercial banks.

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Our sample includes 10,165 commercial banks across 118 countries over the period from 1998 to 2012.

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We group the countries according to their income levels in low-, middle-, and high-income countries.

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The determinants of bank profitability vary quite widely across the different country income categories.

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Also, the recent financial crisis has affected bank profitability, with different effects on the banks in the three country income categories.

Abstract

Using a broad bank-level dataset and the GMM estimator technique described by [Arellano and Bover \(1995\)](#), this paper analyses how bank-specific characteristics, macroeconomic variables, and industry-specific factors affect the profitability of 10,165 commercial banks across 118 countries over the period from 1998 to 2012. Grouping the countries according to three income levels, we show that the determinants of bank profitability included in our model can explain existing profitability differences among commercial banks

in low-, middle-, and high-income countries. The profitability determinants vary quite widely across the different levels of income in terms of significance, sign and size of the effect. The level of income has thus an important impact on the determinants of bank profitability.

Introduction

The role of commercial banks and its development has always attracted the interest of academic research, as it is well known that commercial banks play an important role for the economic development of a country, and that an efficient and profitable banking system is a necessary condition for economic growth. As Beck, Demirguc-Kunt, and Levine (2009) point out, the financial systems and standard indicators of financial intermediary across the world deepened over the past decades along many dimensions. However, progress has been uneven across income groups. Specifically, the deepening has been concentrated in high-income countries, while there has been no significant deepening in middle- and low-income countries. According to this study, various economic and institutional features are different between low-middle- and high-income countries. In this paper, we investigate if and why profitability of commercial banks varies across these income groups and whether bank profitability determinants depend on the countries' income level and their level of economic development, respectively. The main purpose of our paper is thus to better understand the impact of a country's income situation on bank profitability and its main determinants.

For this purpose, we use a large sample with over 10,000 banks from 118 countries around the world over the time period from 1998 to 2012. We analyze which environmental (external) and management-determined (internal) factors have an impact on bank profitability and whether the determinants vary between banks operating in low-, middle and high-income countries. In addition, we analyze the impact of the recent financial crisis on commercial bank profitability within the context of different levels of economic development.

The existing literature on bank profitability is quite large and provides a comprehensive examination of the effects of bank-specific, industry-specific, and macroeconomic determinants on bank profitability. Most of the papers,

however, study this topic within a single-country setup or a small group of countries from either developed or developing countries. The wide range of results from these studies strongly suggests that the income group level has an important impact not only on bank profitability, but also on its determinants.

Only a few papers have analyzed bank performance for a larger sample of countries (e.g., Demirguc-Kunt and Huizinga, 1999, Flamini et al., 2009, Micco et al., 2007). However, none of these studies has explicitly investigated the impact of the income group level and a country's economic development on bank profitability (determinants). Our paper builds on the work by Molyneux and Thornton (1992), Demirguc-Kunt and Huizinga (1999), and Micco et al. (2007). Using methods comparable to those of other scholars (e.g., Athanasoglou et al., 2008, Garcia-Herrero et al., 2009, Staikouras and Wood, 2004), we analyze the effect of bank-specific (e.g., capital ratio, operational efficiency, bank size), industry-specific (e.g., ownership and concentration) and macroeconomic determinants (e.g., inflation and cyclical output) on bank profitability. Economies are divided according to the World Bank's classification based on gross national income (GNI) per capita. By grouping the countries depending on their income level and by separately considering these groups, we are able to analyze how the relevant determinants affect bank profitability and how these effects differ between the different income categories.

With this paper, we want to investigate whether the different development levels and the uneven deepening in the financial intermediation industry over the past decade and the various economic and institutional features across the income groups have an impact on bank profitability and the determinants of bank profitability. In order to do so, we use recent data of more than 10,000 commercial banks across 118 countries and for the longest relevant period (from 1998 to 2012). By applying a dynamic GMM technique, we are able to account for profit persistence and potential endogeneity problems. Former papers also using a multi-country-setup, including the study closest to ours from Demirguc-Kunt and Huizinga (1999), have used linear OLS methods, which however, lead to inconsistent results (e.g., Baltagi, 2001). Finally, we provide new evidence on how the financial crisis has affected commercial banks in different environments of market development. Until now, only a few papers have analyzed the impacts of the recent financial crisis on the

determinants of bank profitability (see, e.g., Beltratti and Stulz, 2011, Dietrich and Wanzenried, 2011). The remainder of the paper is structured as follows. Section 2 surveys the relevant literature on banking profitability. Section 3 outlines our model and the dependent and independent variables used in our analyses. Section 4 describes the data sample and methodology used. Section 5 presents the results of our empirical analysis, and Section 6 concludes.

Section snippets

Review of the literature

Bank performance has been extensively studied in the past. Early work goes back to Short (1979) and Bourke (1989), who were followed by a series of papers which attempted to identify some of the major determinants of bank profitability. The respective empirical studies have focused their analyses either on cross-country evidence or on the banking system of individual countries. The studies of Molyneux and Thornton (1992), Demirguc-Kunt and Huizinga (1999), Staikouras and Wood (2004), Goddard,

Determinants of bank profitability and variable selection

This section describes both the dependent and independent variables considered in our study. Within our presentation of the independent variables, we consider both bank-specific profitability determinants as well as the macroeconomic and industry-specific characteristics.

Data and methodology

This section identifies the sources of our data, presents the data, and describes the regression model that we used to investigate the effects of internal and external factors on bank profitability.

Empirical results

In a first step, we carry out univariate tests in order to identify some basic relationships in our data. The results for our main profitability variable ROAA are in Section 5.2.1. In a second step, we carry our regression analyzes for the

return on equity and the net interest margin, which are our alternative profitability measures; the results can be found in Section 5.2.2.

Conclusions

Banking profitability around the world differs widely as commercial banks have to deal with different macroeconomic environments, different tax policies, and different institutional realities. Applying the GMM estimator technique described by Arellano and Bover (1995) on a large cross-country data set of commercial banks across 118 countries over the period from 1998 to 2012, this paper analyzes how bank characteristics (including ownership), macroeconomic variables and industry-specific

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In addition, profits are likely be pro-cyclical given that economic influences net interest income through lending activity. It is the demand for lending that is increasing (decreasing) in cyclical upswings (downswings) as argued by Dietrich and Wanzenried (2014). Additionally, there is a vast literature that shows that economic growth stimulates the financial system (e.g., Demirguc-Kunt and Huizinga, 1999; Bikker and Hu, 2002; Athanasoglou et al., 2008; Albertazzi and Gambacorta, 2009).

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